Rajeev Jain

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Google Scholar

Principal Software Engineer | ML Infrastructure | HPC | Scientific Computing

Professional Profile

Principal software engineer with 16+ years delivering production systems at the intersection of scientific computing, ML infrastructure, and HPC. Led UXarray from zero to the standard Python tool for DOE labs and NCAR. Ran 10,000+ ML training experiments across Summit, Theta, and Cori for cancer drug response research. Reduced FLASH-X checkpoint overhead 40–70% on Summit via async HDF5 and lossy compression. Generated a 101M-element MONJU reactor mesh on 712 processors in under 7 minutes — a mesh the serial path could not build at all. Two R&D 100 Awards (2022, 2023). Work spans Python platforms, C++ and Fortran HPC codebases, exascale GPU porting, and cross-institution program coordination.

Education

University of Chicago — 2020
M.S. in Computer Science

Arizona State University — 2009
M.S. in Structural Engineering
Advisors: Dr. S.D. Rajan (ASU), Dr. Ashok Belegundu (PSU)
Graduate Fellowship Recipient (2007–2009)

Indian Institute of Technology (ISM) Dhanbad — 2006
B.Tech. in Mechanical Engineering

Professional Experience

Argonne National Laboratory — **2009 – Present**
Principal Specialist, Research Software Engineering — Mathematics and Computer Science Division, Chicago, IL

- Research software engineer across climate analytics, scientific AI, simulation infrastructure, and DOE research software programs.
- Work includes architecture, implementation, testing, release engineering, performance optimization, and cross-institution coordination.
- Major efforts include UXarray, FLASH-X, CANDLE / IMPROVE, Pangu-Weather on Aurora, MeshKit, and urban microclimate software workflows.

University of Chicago — **2023 – Present**
Staff At-Large — Chicago, IL

- Joint appointment supporting cancer pharmacogenomics and Earth science collaborations.
- Mentor students and research staff on software design, experimentation, and reproducibility.

Arizona State University — **2007 – 2009**
Research and Teaching Assistant — Structural and Computational Mechanics Lab, Tempe, AZ

- Conducted FEM-based research in blast mitigation and supported structural analysis coursework.

Earlier Industry Experience — **2005 – 2007**
Project Engineer, Wipro Technologies; Engineering Intern, Tata Motors — India

Research Software Leadership and Projects

UXarray — **2022 – Present**
Lead Developer — Open-source climate analysis platform

- Lead developer for a Python library enabling analysis of native unstructured climate and weather grids.
- Designed the core API, conservative zonal averaging via Gauss-Legendre quadrature, geometry operators, grid I/O for ESMF / MPAS / SCRIP / HEALPix, automated tests, CI/CD, and release workflows.
- Project used by NCAR, DOE labs, and universities. Tutorials and presentations delivered at SC24, AMS, SciPy, EGU, AGU, and related venues.

Pangu-Weather on Aurora **2024 – Present**
Software Engineer AI weather modeling on exascale Intel GPUs

- Reimplemented the Pangu-Weather framework in PyTorch using the Spectral Fourier Neural Operator for Aurora.
- Adapted the workflow to Intel GPUs and ran it at large scale on Aurora.

CANDLE / IMPROVE **2020 – 2025**
Core Contributor Cancer AI infrastructure and benchmarking

- Built hyperparameter optimization and benchmarking infrastructure for cancer drug response prediction.
- Ran 10,000+ model training experiments across Summit, Theta, and Cori, with automated cross-study validation and reproducible workflows.
- Supported a 15+ researcher multi-lab collaboration spanning Argonne, LLNL, and ORNL. Work contributed to an R&D 100 Award in 2023.

FLASH-X **2019 – 2024**
I/O and Compression Lead Multiphysics simulation software

- Implemented asynchronous HDF5 I/O with Argobots and integrated SZ3 / ZFP compression for checkpoint and restart workflows.
- Reduced checkpoint overhead by 40-70% on Summit and achieved storage savings above 50%.
- Enabled cross-checkpoint restart between AMReX and Paramesh solvers. Work contributed to an R&D 100 Award in 2022.

MeshKit **2009 – 2016**
Principal Investigator / Software Lead DOE NEAMS reactor mesh generation

- Principal investigator for MeshKit within DOE NEAMS, leading open-source C++ toolkit for automated nuclear reactor core mesh generation.
- Developed parallel CoreGen workflow: 712 processors, 101 million hexahedral elements, 14 GB MONJU reactor mesh in under 7 minutes — serial path could not run at all.
- Built AssyGen/CoreGen pipeline encoding reactor lattice hierarchy; supported PostBL boundary-layer generation and multi-format mesh workflows.

Urban ECP **2016 – 2019**
Software Lead DOE urban microclimate and CFD workflows

- Coordinated software work across multiple national labs for urban microclimate and CFD workflows, including proposal development and shared workflows.
- Developed integrations linking mesoscale weather (WRF) and high-fidelity CFD (Nek5000) for downtown Chicago urban boundary layer modeling.

Funding and Research Programs

- **DOE SEATS** (active): Software Ecosystem for Advancing Climate Tools and Services.
- **NSF Raijin** (active): collaborative research in climate model analysis.
- **DOE ECP CANDLE** (2017 – 2023): core contributor on cancer AI infrastructure.
- **DOE NEAMS** (2009 – 2016): principal investigator for MeshKit and reactor mesh generation workflows.
- **Urban ECP**: software lead and coordinator for urban microclimate and CFD workflows across multiple labs.

Awards & Recognition

- R&D 100 Award, CANDLE (2023).
- R&D 100 Award, FLASH-X (2022).
- ATPESC Scholar, Argonne Training Program on Extreme-Scale Computing (2015).
- Best Paper Award, International Meshing Roundtable (2010).
- Graduate Fellowship, Arizona State University (2007 – 2009).

Peer-Reviewed Journal Articles

2025 Partin, A., Vasanthakumari, P., Narykov, O., Wilke, A., Koussa, N., Jones, S., Zhu, Y., **Jain, R.**, Overbeek, J. C., Fernando, G. D., Sanchez-Villalobos, C., Garcia-Cardona, C., Mohd-Yusof, J., Chia, N., Wozniak, J. M., Ghosh, S., Pal, R., Brettin, T. S., Weil, M. R., and Stevens, R. L. "Benchmarking community drug response prediction models: datasets, models, tools, and metrics for cross-dataset generalization analysis." *Briefings in Bioinformatics*.

2022 Dubey, A., Weide, K., O'Neal, J., Dhruv, A., Couch, S., Harris, J. A., Klosterman, T., **Jain, R.**, Rudi, J., Messer, B., Pajkos, M., Carlson, J., Chu, R., Wahib, M., Chawdhary, S., Ricker, P. M., Lee, D., Antypas, K., Riley, K. M., Daley, C., Ganapathy, M., Timmes, F. X., Townsley, D. M., Vanella, M., Bachan, J., Rich, P. M., Kumar, S., Endeve, E., Hix, W. R., Mezzacappa, A., and Papatheodore, T. "Flash-X: A multiphysics simulation software instrument." *SoftwareX*.

2022 Harris, J. A., Chu, R., Couch, S. M., Dubey, A., Endeve, E., Georgiadou, A., **Jain, R.**, Kasen, D., Laiu, M. P., Messer, O. E. B., O'Neal, J., Sandoval, M. A., and Weide, K. "Exascale models of stellar explosions: quintessential multi-physics simulation." *International Journal of High Performance Computing Applications*.

2020 **Jain, R.**, Luo, X., Sever, G., Hong, T., and Catlett, C. "Representation and evolution of urban weather boundary conditions in downtown Chicago." *Journal of Building Performance Simulation*.

2018 Wozniak, J. M., **Jain, R.**, Balaprakash, P., Ozik, J., Collier, N. T., Bauer, J., Xia, F., Brettin, T., Stevens, R., Mohd-Yusof, J., Garcia-Cardona, C., Van Essen, B., and Baughman, M. "CANDLE/Supervisor: A workflow framework for machine learning applied to cancer research." *BMC Bioinformatics*.

2014 Mahadevan, V. S., Merzari, E., Tautges, T., **Jain, R.**, Obabko, A., Smith, M., and Fischer, P. "High-resolution coupled physics solvers for analysing fine-scale nuclear reactor design problems." *Philosophical Transactions of the Royal Society A*.

2012 Tautges, T. J. and **Jain, R.** "Creating geometry and mesh models for nuclear reactor core geometries using a lattice hierarchy-based approach." *Engineering with Computers*.

2008 Argod, V., Belegundu, A. D., Aziz, A., Agrawala, V., **Jain, R.**, and Rajan, S. D. "MPI-enabled shape optimization of panels subjected to air blast loading." *International Journal for Simulation and Multidisciplinary Design Optimization*.

Peer-Reviewed Conference and Workshop Proceedings

2025 Gwinn, J., Wozniak, J. M., **Jain, R.**, Zhu, Y., Partin, A., Brettin, T., and Stevens, R. "A Workflow for Error Analysis for Drug Response Prediction via Statistical Standardization and Distribution Analysis." *Workshop on Workflows in Support of Large-Scale Science (WORKS)*.

2024 **Jain, R.**, Tang, H., Dhruv, A., and Byna, S. "Enabling Data Reduction for Flash-X Simulations." *SC24 Workshops: DRBSD-10*.

2024 **Jain, R.**, Wozniak, J. M., Partin, A., Wilke, A., Zhu, Y., Vasanthakumari, P., Narykov, O., Overbeek, J., Weaver, R., Wang, C., Liu, Y., Weil, M. R., Brettin, T., and Stevens, R. "Cross-HPO: Optimizing Neural Networks for Cancer Drug Response Using Hyperparameter Tuning on Multiple Pharmacogenomic Datasets." *Computational Approaches for Cancer Workshop at SC24*.

2023 Wozniak, J. M., **Jain, R.**, Wilke, A., Weaver, R., Partin, A., Brettin, T., and Stevens, R. "An Automation Framework for Comparison of Cancer Response Models Across Configurations." *IEEE e-Science*.

2023 Dhruv, A., **Jain, R.**, O'Neal, J., Weide, K., and Dubey, A. "Framework and Methodology for Verification of a Complex Scientific Simulation Software, Flash-X." *Congress in Computer Science, Computer Engineering, and Applied Computing*.

2022 **Jain, R.**, Tang, H., Dhruv, A., Harris, J. A., and Byna, S. "Accelerating Flash-X Simulations with Asynchronous I/O." *PDSW*.

2021 **Jain, R.**, Shah, A., Mohd-Yusof, J., Wozniak, J. M., Brettin, T., Xia, F., and Stevens, R. "Probing Decision Boundaries in Cancer Data Using Noise Injection and Counterfactual Analysis." *CAFCW at SC21*.

2021 **Jain, R.**, Weide, K., Chawdhary, S., and Klostermann, T. "Checkpoint / Restart for Lagrangian Particle Mesh with AMR in Community Code FLASH-X." *SuperCheck*.

2015 **Jain, R.** and Tautges, T. J. "NEAMS MeshKit: Nuclear Reactor Mesh Generation Solutions." *International Congress on Advances in Nuclear Power Plants*.

2015 **Jain, R.**, Mahadevan, V., and O'Bara, R. "Simplifying Workflow for Reactor Assembly and Full-Core Modeling." *Mathematics and Computation (MC2015)*.

2014 **Jain, R.** and Tautges, T. J. "Generating Unstructured Nuclear Reactor Core Meshes in Parallel." *Procedia Engineering*.

2013 **Jain, R.** and Tautges, T. J. "PostBL: Post-Mesh Boundary Layer Mesh Generation Tool." *International Meshing Roundtable*.

2012 **Jain, R.** and Tautges, T. J. “RGG: Reactor Geometry (and Mesh) Generator.” *International Congress on Advances in Nuclear Power Plants.*

2012 Mohanty, S., **Jain, R.**, Majumdar, S., Tautges, T. J., and Srinivasan, M. “Coupled Field Structural Analysis of HTGR Fuel Brick Using Abaqus.” *International Congress on Advances in Nuclear Power Plants.*

2010 Tautges, T. J. and **Jain, R.** “Creating Geometry and Mesh Models for Nuclear Reactor Core Geometries Using a Lattice Hierarchy-Based Approach.” *19th International Meshing Roundtable.* Best Paper Award.

Conference Abstracts, Tutorials, and Presentations

2025 Eroglu, O., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. M. “UXarray: Extending Xarray to Support the Analysis of Native, Kilometer-Scale Unstructured Grids in Python.” *105th AMS Annual Meeting.*

2025 Clyne, J., Chen, H., Chmielowiec, P., Eroglu, O., Hannay, C., Jacob, R., **Jain, R.**, Medeiros, B., Ullrich, P., and Zarzycki, C. “UXarray: Extending Xarray for Enhanced Support of Unstructured Grids.” *EGU General Assembly.*

2024 **Jain, R.** “Tutorial: UXarray for Analysis of Unstructured Climate Data.” *SC24.*

2024 Chmielowiec, P., Chen, H., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. M. “UXarray: Extending Xarray with Support for Unstructured Grids.” *104th AMS Annual Meeting.*

2024 Eroglu, O., Chen, H., Chmielowiec, P., Clyne, J., DeCiampa, C., Hannay, C., Jacob, R., **Jain, R.**, Loft, R., Medeiros, B., Sun, L., Ullrich, P., and Zarzycki, C. “UXarray: Extensions to Xarray to Support Unstructured Grids.” *EGU General Assembly.*

2024 Clyne, J., Chen, H., Chmielowiec, P., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Ullrich, P. A., and Zarzycki, C. M. “UXarray: An Extension to Xarray Supporting Analysis and Visualization of Kilometer-Scale Model Outputs.” *AGU Fall Meeting.*

2024 **Jain, R.** “UXarray Tutorial.” *ESDS Annual Event.* Video.

2023 **Jain, R.** “UXarray for Unstructured Climate Data.” *SciPy 2023.*

2023 **Jain, R.** “Data Reduction for FLASH-X Simulations.” *HDF User Group 2023.*

2023 Clyne, J., Chen, H., Chmielowiec, P., DeCiampa, C., Eroglu, O., Hannay, C., Jacob, R. L., **Jain, R.**, Medeiros, B., Sun, L., Ullrich, P. A., and Zarzycki, C. M. “UXarray: A Community-Developed Tool for the Analysis and Visualization of Model Output on Unstructured Grids in Python.” *AGU Fall Meeting.*

2023 Jacob, R. L., Ullrich, P. A., Zhang, C., **Jain, R.**, Chen, H., Po-Chedley, S., and Vo, T. “Creating New Standards to Simplify Earth System Model Analysis.” *AGU Fall Meeting.*

2019 Sever, G., Obabko, A., Jacob, R., **Jain, R.**, and Catlett, C. “Simulations of Idealized and Real Urban Boundary Layer Environments Using NWP and CFD Models.” *18th Conference on Mesoscale Processes.*

2019 Jacob, R. L., Sever, G., Obabko, A., **Jain, R.**, and Catlett, C. E. “CFD Simulations of Flow in Realistic Urban Geometries Initialized from Weather Models.” *AGU Fall Meeting.*

2018 Obabko, A., **Jain, R.**, Jacob, R., Paoli, R., Sever, G., Grindeanu, I., and Fischer, P. “Nek5000 Wall-Resolved LES for Urban Geometries.” *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment.*

2018 Silva, M. P., Sharma, A., Budhathoki, M., **Jain, R.**, and Catlett, C. E. “Neighborhood Scale Heat Mitigation Strategies Using Array of Things Data in Chicago.” *AGU Fall Meeting.*

2018 Sever, G., Jacob, R., and **Jain, R.** “Mesoscale to Microscale Weather Simulations over the Chicago Downtown Region.” *10th International Conference on Urban Climate / 14th Symposium on the Urban Environment.*

Technical Reports, Manuals, and Other Publications

2020 Goldring, N., Bruhwiler, D., Nash, B., Wu, Z., Nagler, R., Carter, J., Lerch, J., Suthar, K., Den Hartog, P., **Jain, R.**, and Mahadevan, V. “Multiphysics Design and Optimization of Complex Vacuum Chambers (Phase II Final Technical Report).”

2017 **Jain, R.** “Reactor Geometry (&Mesh) Generator Users Manual.”

2016 **Jain, R.**, Vanderzee, E., Grindeanu, I., and Mahadevan, V. “Mesh Generation and Algorithm Development for NEAMS.”

2015 **Jain, R.**, Vanderzee, E., and Mahadevan, V. “Update on Development of Mesh Generation Algorithms in MeshKit.” ANL/MCS-TM-355.

2015 **Jain, R.** and Mahadevan, V. “Documentation for MeshKit-Reactor Geometry (&mesh) Generator.” ANL/MCS-TM-354.

2015 Mahadevan, V., Grindeanu, I. R., Ray, N., **Jain, R.**, and Wu, D. “SIGMA Release v1.2: Capabilities, Enhancements and Fixes.” ANL/MCS-TM-357.

2015 Merzari, E., Shemon, E. R., Yu, Y., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Solberg, J., Ferencz, R., and Whitesides, R. “Full Core Multiphysics Simulation with Offline Mesh Deformation.” ANL/NE-15/42.

2015 Merzari, E., Shemon, E. R., Yu, Y. Q., Thomas, J. W., Obabko, A., **Jain, R.**, Mahadevan, V., Tautges, T., Solberg, J., Ferencz, R. M., and Whitesides, R. “Multi-physics Demonstration Problem with the SHARP Reactor Simulation Toolkit.” ANL/NE-15/44.

2014 **Jain, R.** “Report on FY11 Extensions to MeshKit and RGG.” ANL/MCS-TM-316.

2014 **Jain, R.** and Mahadevan, V. “2014 MeshKit Release.” ANL/MCS-TM-344.

2014 Tautges, T. J. and **Jain, R.** “Mesh Copy/Move/Merge Tool for Reactor Simulation Applications.” ANL/MCS-TM-343.

2013 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Merzari, E., and Ferencz, R. M. “SHARP Assembly-Scale Multiphysics Demonstration Simulations.” ANL/NE-13/9.

2013 **Jain, R.** and Tautges, T. J. “MeshKit.” ANL/MCS-TM-336.

2012 Tautges, T. J., Fischer, P., Grindeanu, I., **Jain, R.**, Mahadevan, V., Obabko, A., Smith, M. A., Hamilton, S., Clarno, K., and Baird, M. “A Coupled Thermal / Hydraulics – Neutronics – Fuel Performance Analysis of an SFR Fuel Assembly.” Report to the U.S. Department of Energy.

Software and Open Source

UXarray Lead developer for open-source Python tooling for unstructured grid analysis. GitHub and documentation.

FLASH-X I/O and compression lead for a million-line multiphysics simulation code used in astrophysics, combustion, and fluid dynamics.

CANDLE / IMPROVE Engineering support for HPO, benchmarking, and reproducible model workflows for cancer AI research.

Pangu-Weather PyTorch workflow for AI weather modeling adapted to Intel GPUs on Aurora.

MeshKit Principal investigator for reactor geometry and mesh generation tooling. Source.

Professional Service

- SBIR / STTR proposal reviewer for the U.S. Department of Energy.
- Panelist, “Revolutionizing Public Infrastructure: The Impact of AI and Machine Learning,” 5th Infraday Midwest Event.
- Reviewer for the *Journal of Open Research Software* and NumGrid.
- Program committee member, NumGrid 2020.
- Session chair, Computational Geometries session, MC2015.
- Member, American Nuclear Society (2012 – 2016).
- Volunteer, South Side Science Festival, University of Chicago (2023, 2024).

Mentorship

- Mentored students, research aides, and staff on scientific Python, HPC workflows, systems engineering, and reproducible software development.
- Rylie Weaver (Research Aide, 2022 – 2024): IMPROVE project, HPO techniques for cancer drug response prediction.
- Aaron Zedwick (Student, 2023 – 2024): UXarray development, dual mesh routines, and remapping functionality.
- Mark Bartoszek (Systems Administration, 2023): mentoring on systems administration practices at Argonne.